

ISN selection is made by a 32-bit random generator and is created during connection request phase. The number is generated by a 32-bit clock, which has ISN cycle of 4.55 hours. The clock is incremented approximately every 4μs, for which a unique ISN is generated with a cycle period of 4.55 hours.

- A 16-bit window flag, which indicates the number of bytes that the sender can transmit without receiving acknowledgements from the receiver. This field is used by the receiver to inform the sender about the availability of the buffer in the receiver. Based on this information, the sender controls the flow. Flow control is implemented by the window field.
- The urgent pointer field, which is used to alert the receiver about incoming urgent data and indicates the end of the urgent data within the sequence of the transmission of packs. Value in the urgent pointer is valid, provided URG (urgent) flag is set (that is, URG=1). Urgent pointer value defines the end of urgent data and the start of normal data in bytes. For example, if urgent pointer is 0000000001000000, the receiver will understand that the next 64 bytes of data will be urgent.
- Reserved bits (six bits) are for future use. All flags—URG, ACK, SYN, FIN, RST and PSH—together are known as control fields of TCP. PSH flag is set when the receiver is told to pass the data immediately to the application. Otherwise, the receiver will buffer the segments until sufficient storage is made. RST flag is used to abort TCP connection. When this flag is set, the receiver is advised to terminate the connection for an abnormal condition.

The options header can be of maximum 40 bytes and is always a multiple of four bytes. To make it a multiple of four bytes padding may be used, if required. A few important options of TCP pack are listed below.

Maximum segment size (MSS) option: This is used to indicate the maximum size of the segment that the

Source Port (16 bits)	Destination Port (16 bits)
Length (16 bits)	Checksum (16 bits)
Data (Variable)	

Figure 2: UDP headers

sender can accept. This option is used during connection establishment phase. The sender specifies the MSS. This is a 16-bit option field. The largest block that MSS can specify is $2^{16}-1 = 65,535$ bytes. A TCP pack or segment has minimum 20-byte header like that of an IP header. Thus, the largest block size of data that a TCP segment or IP datagram can carry is of $65,535 - 40 \text{ bytes} = 65,495$ bytes. Declaration of MSS during connection setup is made to overrule TCP default MSS. TCP default MSS is 536 bytes.

Window scale option (WSO):

The TCP pack has a header of 16-bit window. Maximum permissible advertised window size is $2^{16}-1=65,535$ bytes. With WSO, the larger advertised window size may be used for which upward scaling up to 2^{14} is permissible. With WSO, scaled advertised window size may, hence, go up to $2^{14} \times 2^{16}-1 = 1.073,725,440$ bytes.

Times stamp option: This is used for calculating round trip delay, which is used for calculating time-out period useful for flow control.

TCP divides data into manageable packets that are easier to deal with. It provides a guaranteed error-free connection. TCP breaks a network terminal (NT) service pack into several packets. If an error occurs, a small packet can be retransmitted instead of the whole pack.

TCP uses a header to establish a virtual connection to ensure successful transmission of packets and completion of transmission of the whole message. Source and destination port fields keep the trace of packets. Packets belonging to various applications, such as e-mails, on-line charts and multiple running browsers, can run concurrently based on the trace of packets. Sharing of an established virtual link is allowed at the same time. Sequence number is used to reassemble the packets at the receiver possibly received out of sequence due to sharing.

The receiver, on receiving a packet, performs check-sum operation to detect errors. If the packet is received error-free, a positive acknowledgement is sent. If packet is received erroneously, a negative acknowledgement is sent.

The transmitter, on getting either a negative acknowledgement within a time known as time-out period (TOP) or not getting any response within TOP, retransmits the earlier packet. This is Automatic Repeat Request Protocol (ARQ) for error control.

Flags, window and urgent fields are used to manage the TCP connection. Flags are used to connect and terminate a connection. Window field tells the transmitter the speed at which to send packets. When receiver buffer is full, the window is set to zero. This is how throughput is regulated by the transmitter sent by the receiver. Urgent field is used to send urgent information. When a receiver gets a packet with urgent field, it processes the packet with priority and acts accordingly, such as aborting transmission.

There are two variations of layer 3: user datagram protocol (UDP) and Internet control message protocol (ICMP). UDP is used when no sequence number is used. A simple example of this situation would be when a message can be put into one datagram. ICMP is simpler than UDP. Here, the message is put into one datagram only. Besides, the message is for the network and, hence, no addressing is required. ICMP is intended for TCP/IP software only.

UDP datagram is used to carry time-sensitive data like voice and video. UDP works in connection-less mode. It has fewer overhead bits, and overhead fields are quite simple. Format of UDP pack is shown in Figure 2.

UDP headers

The 16-bit source port field indicates the address of the process at the sender, where process means the individual process that is in communication with the same process at the receiver.

The 16-bit destination port refers to the address of the port at the receiver